

Lot streaming in a two-stage assembly system

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ABSTRACT

In this paper, we address a scheduling problem belonging to a two-stage assembly system that can also be viewed as a mass customization system. The first stage of this system consists of a set of subassembly machines, each of which produces a component type. These components are then transferred in sublots to Stage 2, where they are assembled into finished products. Stage 2 consists of multiple parallel machines. Each of these machines represents an assembly facility devoted to a product type. We represent this configuration as a $m+n$ system, where there are m parallel machines at Stage 1 and n parallel machines at Stage 2. Lot-attached and subplot-attached setups, and transfer times/costs are encountered on the machines at both the stages. We assume that the subassemblies are transferred in equal-sized sublots to Stage 2. Given a number of lots (jobs), the problem is to determine the number of sublots to use for each lot and the sequence in which to process the lots. We consider two different objective functions, namely, minimize makespan, and minimize the total cost incurred. In view of the fact that both of these problems are NP-hard, we develop two column generation-based heuristic methods and show their efficacy over direct solution of the mixed integer programming formulations of the underlying problems. In fact, the results of our computational investigation on the use of these methods on large-sized problems reveal attainment of almost optimal solutions within a few seconds of CPU time. We also present some managerial insights.

1. Introduction

The work reported in this paper belongs to the field of production control—a systematic execution and monitoring of production-related activities. It entails production planning, resource management, lot sizing, and inventory management. Given demands of various products, it specifies when, how much and where to produce these products, and it manages resources, inventory, and procurement of materials and subassemblies, in the face of dynamic demand. The performance of a manufacturing facility relies on the effectiveness of the production control system implemented.

Lot streaming is a production control technique that can be used to accelerate the flow of a product through a production system by splitting its production lot into sublots, and then, processing the sublots in an overlapping fashion over the machines, thereby reducing work-in-process and cycle-time. In this paper, we explore the use of lot streaming for a two-stage assembly system. Our work has led to the development of new insights, results, and algorithms for the use of lot streaming in a new setting, thereby contributing to its knowledge base. It has also led to the development of effective solution methodologies for an important class of problems (Fern, 2003).

The use of lot streaming in a two-stage assembly system has been addressed by Sarin et al. (2011). They define a two-stage assembly

system in which the first stage consists of parallel subassembly machines each of which produces a component type, and the second stage consists of only one assembly machine that assembles a final product after all of its requisite components are ready at Stage 1. Lot-detached setups are incurred on all the machines at Stage 1 and Stage 2. They have presented a polynomial-time algorithm to obtain optimal subplot sizes for the objective of minimizing the makespan, given a fixed maximum number of sublots, and also for the case when the subplot sizes are assumed to take integer values. Their work is based on the following properties. They show that there exists an optimal solution in which the subplot sizes used for processing the lot on the subassembly and assembly machines are consistent. Moreover, for any two subassembly machines u and v , if $t_{ul} \geq t_v$ and $t_u + p_u U \geq t_v + p_v U$, then subassembly machine v can be eliminated from the problem (where t_j : lot-detached setup of lot j , p_k : processing time of machine k , and U : number of items in a production lot). For any three machines u , v and w , where $t_{ul} > t_v > t_w$ and $p_u < p_v < p_w$, subassembly machine v is dominated by either machine u or w , and hence, does not dictate the makespan. Yao and Sarin (2014) have extended the single-lot problem to multiple-lots. Their work relies on the following properties derived from the single lot problem. First, there exists an optimal schedule in which the sequences of lots are the same on all the machines. Second, for a given sequence of lots, there exist optimal subplot sizes for each lot such that the completion time of each lot is

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minimized. And finally, there exists an optimal schedule in which each lot is split into consistent sublots. They have presented a branch-and-bound-based methodology that relies on three machine-based lower bounds and another based on the relaxation of the original problem, while four dominance rules that afford reduction in branching of the branch-and-bound tree have also been provided. Their proposed solution method outperforms direct solution of a mathematical model of the problem by CPLEX for both small- and large-sized problem instances. Nejati et al. (2016) have presented a mixed integer, non-linear programming model and a meta-heuristics-based solution method (namely, simulated annealing and genetic algorithm) for a two-stage assembly hybrid flow shop. They have considered multiple lots, multiple machines at Stage 2, sequence-dependent setup time, consistent subplot sizes, and the objective of minimizing the sum of weighted completion times. Our problem is different because we consider lot and subplot-attached setups, multiple lots, multiple machines at Stage 2, and also, a cost-based objective function. However, we assume equal-sized sublots.

Another machine configuration that is similar to the assembly system considered here and to which lot streaming has been applied is hybrid flow shop (Tsubone et al., 1996; Zhang et al., 2003; 2005; and Liu, 2008). Tsubone et al. (1996) have considered a $1+m$ problem (with one machine at Stage 1 and m machines at Stage 2) and have used a simulation model to study the impact of lot sizing from the viewpoint of optimizing the total flow time, makespan, capacity utilization, and the maximum work-in-process level. Zhang et al. (2003, 2005) and Liu (2008) have considered a $m+1$ problem (with m machines at Stage 1 and one machine at Stage 2). Zhang et al. (2003) assume one of the stages to be a bottleneck and subplot sizes to be integer. The problem is, then, formulated as a mixed integer linear programming model, and two heuristic methods are proposed that allocate the sublots as evenly as possible to the machines at Stage 1, which is shown to produce near-optimal solutions. Zhang et al. (2005) assume equal subplot sizes and the objective of minimizing mean completion time. A good lower bound and two heuristic methods are presented. Liu (2008) assumes a given number of sublots, equal subplot sizes, and the objective of minimizing the makespan. Cheng et al. (2016) have presented closed-form expressions to obtain optimal continuous subplot sizes for a single-lot $1+2$ hybrid flow shop problem. Mathematical modeling-based heuristics have also been provided to obtain integer subplot sizes. Cheng and Sarin (2013) have also considered a multiple-lot problem for a $1+2$ hybrid flow shop. A mathematical programming model and some local search heuristic methods have been proposed to minimize makespan and total completion times. Zhang et al. (2017) have considered a multiple-stage hybrid flow shop, which consists of at least one identical machine in parallel at each stage. They have assumed equal subplot sizes without setup or transportation times, and they present a mathematical model and use of a meta-heuristics to minimize the total flow time. Wang et al. (2019) have considered a two-stage hybrid flow shop problem (with sequence-dependent setups) that is motivated by a real-life example at a coating company. They present a mixed-integer linear program for the problem and use of heuristic method to minimize the total weighted completion time. Li et al. (2020) have solved a hybrid flow shop lot-streaming problem where the subplot sizes are assumed to be variable. Heuristic methods are presented to minimize four objectives, namely, the penalty caused by the average sojourn time, the energy consumption at the last stage, and the earliness and tardiness values.

1.1. Difference between a two-stage assembly system and hybrid flow shop

The two-stage assembly system that we consider is different from a hybrid flow shop. In a hybrid flow shop, the jobs are processed on the machines sequentially as shown in Fig. 1a, while in a two-stage assembly system, Stage 1 consists of suppliers, each of which produces a type of a

subassembly (or a component). Then, these subassemblies are put together on an assembly machine at Stage 2 (see Fig. 1b). A real-life instance of such a system includes the dressing of engines and transmissions by suppliers (Stage 1) for their assembly at an assembly plant (Stage 2), and the fabrication of integrated circuits and manufacturing of power supplies, among other components, by suppliers (at Stage 1) for their assembly on printed circuit boards (at Stage 2).

1.2. A numerical example of lot streaming in a two-stage assembly system

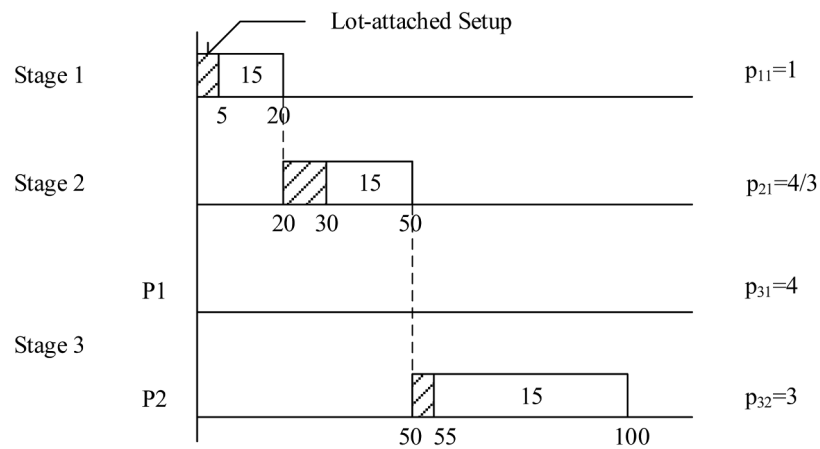
Figs. 2 and 3 depict a numerical example of a two-stage assembly system without and with the use of lot streaming, respectively. The first stage consists of five machines, each of which produces subassemblies, namely, CPU, memory, storage, camera and accessories. They are then transferred to one of the assembly locations (United States/Asia) at Stage 2. An attached setup time is incurred before the processing of each lot and its sublots on the machines at both the stages, and a lot-attached transfer time is incurred from Stage 1 to Stage 2. Three different lots of product types with each product type requiring different types of subassemblies, are to be assembled on the locations at Stage 2. The products in lots 1, 2, and 3 require CPU, memory, camera, and accessories; CPU, memory, storage, and camera; and CPU, memory, storage, and accessories, respectively. Note that the makespan value reduces to 187.5 units when lot streaming is used (it splits lot 1 into 3 equal-sized sublots and lot 2 and 3 into 2 equal-sized sublots) from its value of 197.5 units when lot streaming is not used.

1.3. Importance of the problem addressed

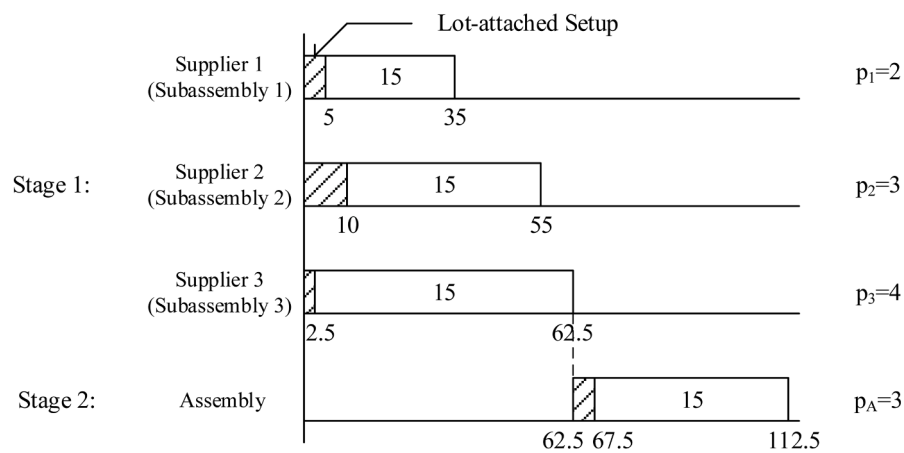
Nowadays, it is imperative for manufacturing companies to run operations efficiently in order to stay competitive in the global market place. The two-stage assembly configuration that we consider in this paper can be viewed as a mass customization system, which has emerged as an important production mode for industry to achieve lower cost, shorter cycle time, and greater product variety (See Pine et al., 1993). The concept of mass customization originated in late 1980s and has been effectively used in apparel and computer industries, among others. Mass customization promotes the idea of postponing the final assembly step, which differentiates one product type from another until the very last stage of production. The best-known practice of mass customization is by Dell Computer (Fern, 2003), where a customer picks his/her computer processor, memory, storage, and other equipment, on Dell's website. Dell's supply chain is configured to obtain subassemblies from suppliers (Stage 1), and then, to assemble the requisite systems in different market areas (Stage 2) to reduce operating costs while improving responsiveness to customer demand. A literature review on mass customization and a framework for future research direction can be found in Silveira et al. (2001) and Fogliatto et al. (2012).

The key decisions that are pertinent for this environment pertain to scheduling of the lots for processing by the vendors at the first stage, and also, for their assembly at the second stage. Because of the customization of products and dynamic nature of their demands, vendors cannot produce necessary components (subassemblies) ahead of time, and production only starts upon the arrival of customer orders. Lot streaming is directly applicable to this situation. It specifies the number and sizes of the sublots for each lot as well as a sequence for processing the lots of different products to accelerate production while at the same time minimizing the total cost incurred (due to setups, transfers, and inventory holding, among others.) A similar situation arises in supply chain management, and also, in cloud manufacturing where a task is processed in an overlapping fashion over shared resources, and therefore, demands it's splitting like that of determining sublots.

A review of work reported on lot streaming is given in Chang and Chiu (2005) and Cheng et al. (2013). We do not provide a review of this work here for the sake of brevity. The interested reader is referred to these articles, especially the one by Cheng et al. (2013), which presents



(a) Hybrid Flow Shop. Lot size = 15, and p_{ij} =unit processing times on machine j at Stage i .



(b) Two-stage Assembly System. Lot size = 15, and p_j =unit processing time on machine j of Stage 1, p_A =unit processing time on the assembly machine at Stage 2.

Fig. 1. A hybrid flow shop and a two-stage assembly system.

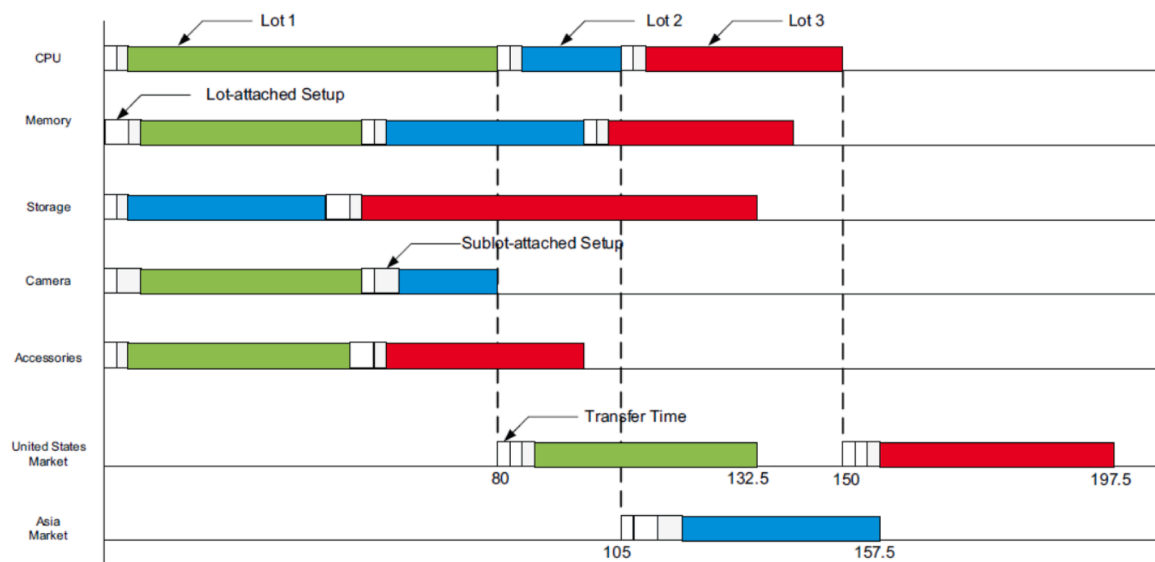


Fig. 2. Processing of multiple lots without lot streaming in a two-stage assembly system.

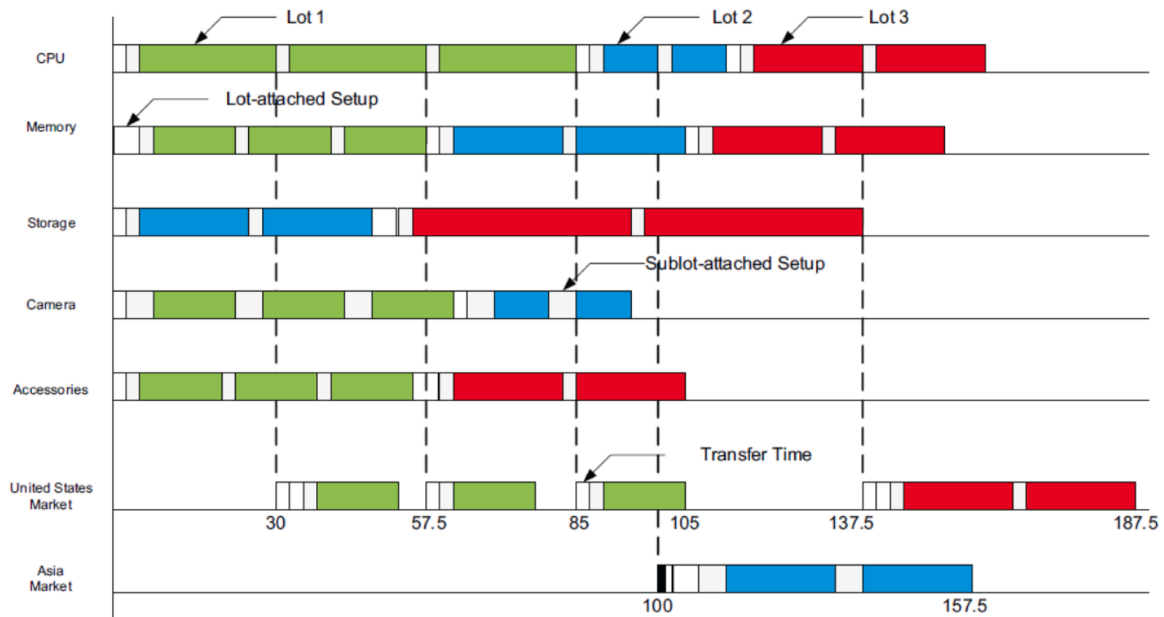


Fig. 3. Processing of multiple lots with lot streaming in a two-stage assembly system.

the variety of lot streaming problems addressed in the literature and the contributions made for their solution. Most of this work has been devoted to flow shops and the time-based objective functions pertaining to minimizing makespan and flow time. However, as indicated in [Section 1.1](#), our problem is different from flow shop scheduling because of both the machine configuration and the objective function considered, which makes it a very complex problem, and it has not been addressed in the literature. We propose a column generation method for the solution of this problem that decomposes it into two problems, called master problem and subproblem, by splitting the decision variables into two subsets. We use two different versions of this decomposition scheme. In one decomposition scheme, the number of sublots and sublot sizes are determined in master problem and the sequence in which to process the lots is determined in the subproblem, while the roles are reversed in the second version. We present results of our experimental investigation to demonstrate the effectiveness of our proposed methodologies over solving the entire problem at once.

1.4. Problem statement

The problem that we address can be defined as follows: Given a set of lots, each consisting of a known number of products of a product type, and a maximum number of sublots in which to divide each lot, split each lot into equal-sized sublots, and sequence the lots for processing in a two-stage assembly system that incurs an attached setup time before the processing of each lot and its sublots on the machines at both the stages and also a sublot-attached transfer time from Stage 1 to Stage 2, so as to optimize a performance measure, for which we consider both minimization of makespan and total cost. Accordingly, we designate this problem as TSAS-MP (two-stage, assembly system, makespan problem) and TSAS-CP (two-stage, assembly system, cost-based problem).

The rest of this paper is organized as follows. In [Section 2](#), we present the problem in detail and introduce model formulations for the TSAS-MP and TSAS-CP, respectively. In [Section 3](#), two column generation methods are presented. Results of our computational investigation on the use of these methods are presented in [Section 4](#). Some managerial insights are presented in [Section 5](#), and finally, concluding remarks are made in [Section 6](#).

2. Problem description and formulation

2.1. Problem description

The TSAS-MP and TSAS-CP that we consider consist of a set of lots J that is to be processed in a two-stage assembly system. Each lot $j \in J$ consists of Q_j products, and each product of lot j requires S_j subassemblies that are produced by S_j vendors (each vendor is represented as a machine). Each vendor supplies one subassembly type. Each lot is split into equal-sized sublots, and a requisite number of subassemblies for the products of each sublot of lot j are then transferred to assembly machine a_j at Stage 2 for the formation of their assemblies. Attached setup times are incurred before the processing of each lot and its sublots and for the transfer of sublots from Stage 1 to Stage 2. We consider the problem for two different objectives of minimizing the makespan and minimizing the total cost pertaining to makespan, setup of lots and sublots, transfer of sublots, tardiness, and inventory holding. The problem features are summarized below:

Production Configuration

- 1) There are multiple subassembly machines operating at the first stage.
- 2) There are one or more assembly machines at the second stage.
- 3) Each lot can be split into multiple equal-sized sublots with the number of sublots not exceeding a maximum number of sublots allowed.
- 4) Attached setup times are incurred before the processing of a lot and sublot at each stage, and for the transfer of the sublots from Stage 1 to Stage 2 (See [Fig. 3](#)).
- 5) Products of each lot are pre-assigned to an assembly machine at Stage 2, which links each product to its demand for the assembly operation.

Objective

- 1) Minimize the makespan;
- 2) Minimize total cost, which comprises makespan cost, setup cost, transfer cost, tardiness cost, and inventory holding cost.

Decisions

- 1) Sequence in which to process the lots;

- 2) Number of sublots for each lot;
- 3) Size of each subplot.

2.2. Notation and assumptions

In this section, we present the notation that we use in this paper. Also, we list the assumptions that we have made.

Sets

- S : Set of subassembly machines at Stage 1
- J : Set of lots to be scheduled
- J^k : Set of lots that require subassembly machine k (at Stage 1), $k \in S$
- S_j : Set of subassembly machines required by lot j , $\forall j \in J$, i.e., the set of subassemblies required by lot j
- A : Set of assembly machines at Stage 2
- A^k : Set of lots that are to be assembled on machine k (at Stage 2), $k \in A$.
- Parameters
- a_j : Machine where the products of lot j are assembled, $\forall j \in J$
- Q_j : Number of products in lot j , $\forall j \in J$
- n_j : Maximum number of sublots for lot j , $\forall j \in J$
- ls_j^k : Lot-attached setup time for lot j on subassembly machine k , $\forall k \in S_j, j \in J$
- la_j : Lot-attached setup time for lot j on assembly machine A_j , $\forall j \in J$
- ps_j^k : Processing time per subassembly for each product of lot j on subassembly machine k at the first stage, $\forall k \in S_j, j \in J$
- pa_j : Processing time per product of lot j on assembly machine A_j at the second stage, $\forall j \in J$
- ts_j^k : Sublot-attached setup time prior to the processing of subassemblies for a subplot of lot j on subassembly machine k at the first stage, $\forall k \in S_j, j \in J$
- ta_j : Sublot-attached setup time prior to assembling the sublots of lot j on assembly machine A_j at the second stage, $\forall j \in J$
- tf_j^k : Sublot-attached transfer time for the sublots of lot j from the subassembly machine k , $\forall k \in S_j$, to the assembly machine a_j , $j \in J$
- M : A large positive number
- ϵ : A small positive number used as a tolerance for constraint violation
- Variables

$$x_{ij} = \begin{cases} 1, & \text{if lot } i \text{ is processed before lot } j, \\ 0, & \text{otherwise, } \forall i \in J, j \in J, i \neq j; \end{cases}$$

$$z_{ju} = \begin{cases} 1, & \text{if subplot } u \text{ of lot } j \text{ is used,} \\ 0, & \text{otherwise, } \forall u = 1, \dots, n_j, j \in J \end{cases}$$

- s_{ju} : Size of subplot u of lot j , $\forall u = 1, \dots, n_j, j \in J$
- CS_{ju}^k : Completion time of subplot u of lot j on subassembly machine k , $u = 0, \dots, n_j, k \in S_j, j \in J$; a dummy subplot $u = 0$ is used to capture the start of a lot on a machine at Stage 1 and Stage 2
- $CA_{ju}^{a_j}$: Completion time of subplot u of lot j on assembly machine a_j , $u = 0, \dots, n_j, j \in J$
- $C_{\max}$: Makespan
- Note that CS_{j0}^k and $CA_{j0}^{a_j}$ are the start times of lot j on subassembly machine k and assembly machine a_j , $\forall k \in S_j, j \in J$, respectively
- In addition, for the cost-based model, we use the following parameters.
- d_j : Due date for lot j , $\forall j \in J$;
- c_1 : Makespan cost per unit time;
- c_2 : Sublot-attached setup cost per unit time for the subassembly machines at Stage 1;
- c_3 : Sublot-attached setup cost per unit time for the assembly machines at Stage 2;

- c_4 : Lot-attached setup cost per unit time for the subassembly machines at Stage 1;
- c_5 : Lot-attached setup cost per unit time for the assembly machines at Stage 2;
- c_6 : Transfer cost per unit time between the machines at Stage 1 and Stage 2;
- c_7 : Tardiness cost per unit time
- c_8 : Inventory cost per unit time

Assumptions

- The products contained in a lot are identical
- The machines at the first and second stages are available at time zero and remain continuously available
- A machine can process only one subplot at any time
- The setup times, transfer times and unit processing times are known in advance
- The processing of a subplot i on an assembly machine can begin only after all the necessary subassemblies have finished processing at the first stage
- Unlimited buffers are available on the machines at the first and second stages

2.3. Model formulation

Model TSAS-MP-MIP

Objective Function

$$\text{Minimize } C_{\max}, \quad (1)$$

Constraints

Makespan Constraint:

$$C_{\max} \geq CA_{mj}^{a_j}, \quad \forall j \in J \quad (2)$$

Stage 1 Subassembly Machine Constraints:

$$CS_{j0}^k \geq CS_{in_i}^k + (x_{ij} - 1) \cdot M, \quad \forall i \neq j, i, j \in J^k, k \in S \quad (3)$$

$$CS_{j1}^k \geq CS_{j0}^k + ls_j^k + ts_j^k + ps_j^k \cdot s_{j1}, \quad \forall k \in S_j, j \in J \quad (4)$$

$$CS_{ju}^k \geq CS_{ju-1}^k + ts_j^k \cdot z_{ju} + ps_j^k \cdot s_{ju}, \quad \forall k \in S_j, u = 2, \dots, n_j, j \in J \quad (5)$$

Stage 2 Assembly Machine Constraints:

$$CA_{j0}^{a_j} \geq CA_{in_i}^{a_j} + (x_{ij} - 1) \cdot M, \quad \forall i \neq j, i, j \in A^k, k \in A \quad (6)$$

$$CA_{j1}^{a_j} = CA_{j0}^{a_j} + la_j + ta_j + pa_j \cdot s_{j1}, \quad \forall j \in J \quad (7)$$

$$CA_{ju}^{a_j} = CA_{ju-1}^{a_j} + ta_j \cdot z_{ju} + pa_j \cdot s_{ju}, \quad \forall u = 2, \dots, n_j, j \in J \quad (8)$$

Transfer Constraints:

$$CA_{j1}^{a_j} \geq CS_{j1}^k + tf_j^k + la_j + ta_j + pa_j \cdot s_{j1}, \quad \forall k \in S_j, j \in J \quad (9)$$

$$CA_{ju}^{a_j} \geq CS_{ju}^k + (ta_j + tf_j^k) \cdot z_{ju} + pa_j \cdot s_{ju}, \quad \forall k \in S_j, u = 2, \dots, n_j, j \in J \quad (10)$$

Sequence Constraints:

$$x_{ij} + x_{ji} = 1, \quad \forall i, j \in J, i \neq j \quad (11)$$

$$x_{ij} + x_{jl} + x_{li} \leq 2, \quad \forall i, j, l \in J, i \neq j \neq l \quad (12)$$

Sublot Formation Constraints:

$$\sum_{u=1}^{n_j} s_{ju} = Q_j, \quad \forall j \in J \quad (13)$$

$$s_{ju} \leq z_{ju} \cdot Q_j, \quad \forall u = 1, \dots, n_j, j \in J \quad (14)$$

$$s_{ju} \geq s_{j,u+1}, \quad \forall u = 1, \dots, n_j - 1, j \in J \quad (15)$$

Sublot Size Constraints:

$$s_{ju} - s_{j1} \geq Q_j(z_{ju} - 1), \quad \forall u = 1, \dots, n_j, j \in J \quad (16)$$

$$s_{ju} - s_{j1} \leq Q_j(1 - z_{ju}), \quad \forall u = 1, \dots, n_j, j \in J \quad (17)$$

$$s_{ju} \leq Q_j \cdot z_{ju}, \quad \forall u = 1, \dots, n_j, j \in J \quad (18)$$

$$s_{ju} - s_{j1} \leq z_{ju} - \varepsilon, \quad \forall u = 1, \dots, n_j, j \in J \quad (19)$$

$$s_{ju} \geq \varepsilon - (1 - z_{ju}), \quad \forall u = 1, \dots, n_j, j \in J \quad (20)$$

Range of Variables:

$$C_{\max} \geq 0, C_{ju}^k \geq 0, CA_{ju}^{aj} \geq 0 \quad \forall k \in S_j, u = 0, \dots, n_j, j \in J,$$

$$x_{ij} \in \{0, 1\}, \quad \forall i, j \in J, i \neq j$$

$$z_{ju} \in \{0, 1\}, \quad \forall u = 1, \dots, n_j, j \in J$$

$$s_{ju} \geq 0, \quad \forall u = 1, \dots, n_j, j \in J$$

Explanation of constraints:

- 1 Makespan constraint: The makespan of the system is greater than or equal to the completion time of the last lot.
- 2 Stage 1 (Subassembly) and Stage 2 (Assembly) Machine Constraints: Constraint sets (3) and (6) guarantee that a lot j can start on a subassembly machine and assembly machine, respectively, only after the previous lots have finished processing on those machines. We specify later a reasonable value of M to be used in these constraints. Constraints (4), (5), (7), and (8) ensure that a subplot u of lot j can start only after subplot $u - 1$ has finished processing. Note that Constraints (4) and (7) determine the start times of the first subplot at Stage 1 and Stage 2, respectively, and are different from Constraints (5) and (8) because of the presence of lot-attached setup times.
- 3 Transfer Constraints: Constraint sets (9) and (10) assert that subplot 1 and subplot $u \in \{2, \dots, n_j\}$ of lot j can start processing only after all needed subassemblies have been completed and transferred to assembly machine a_j . Note that Constraint (9) is different from (10) because of the presence of lot-attached setup time.
- 4 Sequence Constraints: Constraint sets (11) capture the fact that either lot i is processed before lot j or lot j is processed before lot i . Constraint sets (12) are the subtour elimination constraints that enforce that if lot i precedes lot j , lot j precedes lot l , then lot l cannot precede lot i . (see Sarin et al. (2005))
- 5 Sublot Formation Constraints: Constraint set (13) ensures that the summation of all the subplot sizes of a lot is equal to the total number of items (products) of that lot, while Constraint set (14) enforces the size of a subplot to be positive only if it is formed. In case not all the sublots of a lot are formed, without loss of generality, we enforce the highest numbered sublots to take a value of zero. This is assured by Constraint set (15).
- 6 Sublot Size Constraints: Constraint sets (16) - (20) capture the relationships between the sizes of sublots, $s_{ju} \forall u = 1, \dots, n_j, j \in J$, and their indicator variables, $z_{ju}, \forall u = 1, \dots, n_j, j \in J$. These constraints guarantee that if $s_{ju} \geq 0$, then $z_{ju} = 1$, and if $z_{ju} = 1$, then $s_{ju} > 0$. Note that we assume equal subplot sizes at the beginning, and we use s_{j1} as the size of the first subplot, and let the size of a subplot to be less than or equal to the previous subplot. Therefore, by minimizing the size of the first subplot and using indicator variables z_{ju} , the above formulation captures the equality of all the sublots that are used. We have the following four instances:
 - (a) If $z_{ju} = 1$, the constraint sets (16) and (17) enforce $s_{ju} = s_{j1}$, and the constraint sets (18), (19), and (20) hold.

- (b) If $z_{ju} = 0$, the constraint sets (18) and $s_{ju} \geq 0$ guarantee $s_{ju} = 0$, and the constraint sets (16), (17), (19), and (20) hold.
- (c) If $s_{ju} = s_{j1}$, the constraint sets (18) assert $z_{ju} = 1$, and constraint sets (16), (17), (18), and (20) hold.
- (d) If $s_{ju} = 0$, the constraint set (20) enforces $z_{ju} = 0$, and constraint sets (16), (17), (18), and (19) hold.

The TSAS-CP-MIP model is different because of the objective function, which is now cost-based, and it is as follows:

Model TSAS-CP-MIP

Minimize

$$\begin{aligned} & c_1 \cdot C_{\max} + c_2 \cdot \sum_{j \in J} \sum_{k \in S_j} \sum_{u=1}^{n_j} t s_j^k z_{ju} + c_3 \cdot \sum_{j \in J} \sum_{u=1}^{n_j} t a_j z_{ju} + c_4 \cdot \sum_{j \in J} \sum_{k \in S_j} l s_j^k \\ & + c_5 \cdot \sum_{j \in J} l a_j + c_6 \cdot \sum_{j \in J} \sum_{k \in S_j} \sum_{u=1}^{n_j} t_j^k z_{ju} + c_7 \cdot \sum_{j \in J} \max \left\{ 0, CA_{jn_j}^{aj} - d_j \right\} \\ & + c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} Q_j \cdot \left(CA_{jn_j}^{aj} - CS_{j0}^k - l s_j^k - \left(\sum_{u=1}^{n_j} z_{ju} + 1 \right) t s_j^k \right) / 2 - p s_j^k (Q_j + s_{j1}) / 2 \end{aligned} \quad (21)$$

Subject to Constraints (2)–(20).

The first seven terms are straight-forward and they represent makespan cost, the subplot-attached setup cost incurred on subassembly and assembly machines, the lot-attached setup cost incurred on subassembly and assembly machines, transfer cost from Stage 1 to Stage 2, and tardiness cost, respectively. The last term captures the inventory holding cost accrued because of subassemblies in the system, and it is developed as follows:

$$\begin{aligned} & = c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} [s_{j1} \cdot (CA_{jn_j}^{aj} - CS_{j1}^k) \cdot z_{j1} + s_{j1} \cdot (CA_{jn_j}^{aj} - CS_{j2}^k) \cdot z_{j2} + \dots \\ & \quad + s_{j1} \cdot (CA_{jn_j}^{aj} - CS_{jn_j}^k) \cdot z_{jn_j}] \\ & = c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} \left[\sum_{u=1}^{n_j} z_{ju} \cdot s_{j1} \cdot CA_{jn_j}^{aj} - \sum_{u=1}^{n_j} z_{ju} \cdot s_{j1} \cdot CS_{jn_j}^k \right] \\ & = c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} [Q_j \cdot CA_{jn_j}^{aj} - s_{j1} \cdot (CS_{j0}^k + l s_j^k + t s_j^k + p s_j^k \cdot s_{j1}) \\ & \quad + (CS_{j0}^k + l s_j^k + 2 t s_j^k + 2 p s_j^k \cdot s_{j1}) + \dots \\ & \quad + (CS_{j0}^k + l s_j^k + t s_j^k \cdot \sum_{u=1}^{n_j} z_{ju} + p s_j^k \cdot s_{j1} \cdot \sum_{u=1}^{n_j} z_{ju})] \\ & = c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} [Q_j \cdot CA_{jn_j}^{aj} - [s_{j1} \cdot CS_{j0}^k \cdot \sum_{u=1}^{n_j} z_{ju} + s_{j1} \cdot l s_j^k \cdot \sum_{u=1}^{n_j} z_{ju} \\ & \quad + \frac{\left(\sum_{u=1}^{n_j} z_{ju} + 1 \right) \cdot \sum_{u=1}^{n_j} z_{ju} \cdot s_{j1}}{2} \cdot t s_j^k + \frac{\left(\sum_{u=1}^{n_j} z_{ju} + 1 \right) \cdot \sum_{u=1}^{n_j} z_{ju} \cdot s_{j1}^2}{2} \cdot p s_j^k] \\ & = c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} [Q_j \cdot CA_{jn_j}^{aj} \\ & \quad - \left[Q_j \cdot CS_{j0}^k + Q_j \cdot l s_j^k + \frac{Q_j}{2} \cdot t s_j^k \cdot \left(\sum_{u=1}^{n_j} z_{ju} + 1 \right) + p s_j^k \cdot \frac{Q_j (Q_j + s_{j1})}{2} \right] \\ & = c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} Q_j \cdot \left(CA_{jn_j}^{aj} - CS_{j0}^k - l s_j^k - \left(\sum_{u=1}^{n_j} z_{ju} + 1 \right) t s_j^k / 2 - p s_j^k (Q_j + s_{j1}) / 2 \right) \end{aligned}$$

Note that both the models TSAS-MP-MIP and TSAS-CP-MIP contain a large positive number M . We determine a reasonable value of M as follows: Let,

$$M'_k = \sum_{j \in J^k} (ps_j^k \cdot Q_j + ts_j^k \cdot n_j + ls_j^k), \quad \forall k \in S \quad (22)$$

which is the completion time of subassembly station k . Then, an upper bound of M ,

$$\hat{M} = \max_{k \in S} M'_k + \max_{m \in A} \left[\sum_{j \in A^m} \left[Q_j \cdot pa_j + la_j + n_j \cdot \left(ta_j + \sum_{l \in S_j} t_l^j \right) \right] \right]. \quad (23)$$

Note that $\hat{M}$ consists of the maximum of the completion times for all the lots at Stage 1 plus the maximum of the sum of the processing time, transfer time, and setup times for all the lots over all the assembly stations.

2.4. Basic Properties

Property 1. Both the TSAS-MP and TSAS-CP are NP-hard.

Gupta (1988) has shown that if one of the two stages of a two-stage hybrid flow shop has more than one machine, then the problem is NP-hard. This problem is a special case of our two-stage assembly problem. Hence, our problem is NP-hard.

Property 2. There exists an optimal schedule in which the sequence between any pair of lots is identical on the subassembly and assembly machines for the TSAS-MP.

It is easy to see that if the lots are not processed in the same sequence on the subassembly and assembly machines, then we can modify the sequences and make them identical without worsening the makespan. Note that Property 2 need not hold true for the TSAP-CP. However, for the sake of convenience, we consider such sequences for this problem as well.

3. Column generation-based method

In this section, we introduce two column generation-based methods to solve models TSAS-MP-MIP and TSAS-CP-MIP that rely on a two-stage decomposition scheme. The Stage I problem (master problem) is solved as a linear program to generate dual variables for use at Stage II, which generates columns to pass on to Stage I. The subproblem generates columns until a specified termination criterion is met.

3.1. Column generation method 1 (TSAS-MP-CG1 and TSAS-CP-CG1)

In this decomposition scheme, the subproblem determines a sequence among the lots for passing on to the master problem, which then determines subplot sizes for each lot. Consider the following notation.

Definitions:

- ξ : A set of all generated columns.
- x_{ij}^p : p^{th} column of variable x_{ij} , $\forall i \in J, j \in J, i \neq j, p \in \xi$.
- $\lambda_p = \begin{cases} 1, & \text{if column } p \text{ is selected;} \\ 0, & \text{otherwise.} \end{cases} \forall p \in \xi$

Master problem:

The master problem (MP-TSAS-MP-CG1) is as follows:

MP-TSAS-MP-CG1

Minimize

$$C_{\max}$$

subject to:

$$\sum_{p \in \xi} \lambda_p = 1 \quad (24)$$

$$C_{\max} \geq CA_{jnj}^{aj}, \quad \forall j \in J \quad (25)$$

$$CS_{j0}^k \geq CS_{in_i}^k + \left(\sum_{p \in \xi} \lambda_p x_{ij}^p - 1 \right) \cdot \hat{M}, \quad \forall i \neq j, i, j \in J^k, k \in S \quad (26)$$

$$CS_{j1}^k \geq CS_{j0}^k + ls_j^k + ts_j^k + ps_j^k \cdot s_{j1}, \quad \forall k \in S_j, j \in J \quad (27)$$

$$CS_{ju}^k \geq CS_{ju-1}^k + ts_j^k \cdot z_{ju} + ps_j^k \cdot s_{ju}, \quad \forall k \in S_j, u = 2, \dots, n_j, j \in J \quad (28)$$

$$CA_{j0}^{aj} \geq CA_{in_i}^{aj} + \left(\sum_{p \in \xi} \lambda_p x_{ij}^p - 1 \right) \cdot \hat{M}, \quad \forall i \neq j, i, j \in A^k, k \in A \forall i \neq j, i, j \in A^k, k \in A \quad (29)$$

$$CA_{j1}^{aj} = CA_{j0}^{aj} + la_j + ta_j + pa_j \cdot s_{j1}, \quad \forall j \in J \quad (30)$$

$$CA_{ju}^{aj} = CA_{ju-1}^{aj} + ta_j \cdot z_{ju} + pa_j \cdot s_{ju}, \quad \forall u = 2, \dots, n_j, j \in J \quad (31)$$

$$CA_{j1}^{aj} \geq CS_{j1}^k + t_j^k + la_j + ta_j + pa_j \cdot s_{j1}, \quad \forall k \in S_j, j \in J \quad (32)$$

$$CA_{ju}^{aj} \geq CS_{ju}^k + \left(ta_j + t_j^k \right) \cdot z_{ju} + pa_j \cdot s_{ju}, \quad \forall k \in S_j, u = 2, \dots, n_j, j \in J \quad (33)$$

$$\sum_{u=1}^{n_j} s_{ju} = Q_j, \quad \forall j \in J \quad (34)$$

$$s_{ju} \leq z_{ju} \cdot Q_j, \quad \forall u = 1, \dots, n_j, j \in J \quad (35)$$

$$s_{ju} \geq s_{j,u+1}, \quad \forall u = 1, \dots, n_j - 1, j \in J \quad (36)$$

$$s_{ju} - s_{j1} \geq Q_j (z_{ju} - 1), \quad \forall u = 1, \dots, n_j, j \in J \quad (37)$$

$$s_{ju} - s_{j1} \leq Q_j (1 - z_{ju}), \quad \forall u = 1, \dots, n_j, j \in J \quad (38)$$

$$s_{ju} \leq Q_j \cdot z_{ju}, \quad \forall u = 1, \dots, n_j, j \in J \quad (39)$$

$$s_{ju} - s_{j1} \leq z_{ju} - \epsilon, \quad \forall u = 1, \dots, n_j, j \in J \quad (40)$$

$$s_{ju} \geq \epsilon - \epsilon (1 - z_{ju}), \quad \forall u = 1, \dots, n_j, j \in J \quad (41)$$

Subproblem:

Let π , ϕ_{ij}^k , and φ_{ij} be the dual variables associated with Constraint sets (24), (26), and (29). We have the following subproblem. The objective function is the reduced cost of the λ variable.

SP-TSAS-MP-CG1

Minimize

$$\sum_{i \in J} \sum_{j \in J, j \neq i} \sum_{k \in S_j} \hat{M} \cdot \left(\phi_{ij}^k + \varphi_{ij} \right) \cdot x_{ij} - \pi$$

Subject to:

$$x_{ij} + x_{ji} = 1, \quad \forall i \neq j, i, j \in J$$

$$x_{ij} + x_{jl} + x_{li} \leq 2, \quad \forall i \neq j \neq l, i, j, l \in J$$

$$x_{ij} \in \{0, 1\}, \quad \forall i \neq j, i, j \in J$$

If the reduced cost obtained from the SP-TSAS-MP-CG1 is negative, we add the corresponding column, designated x_{ij}^p to the master problem (MP-TSAS-MP-CG1), and then, re-solve MP-TSAS-MP-CG1 to generate a dual solution. This process is continued until the reduced cost becomes non-negative or the maximum allowable number of iterations is reached.

TSAS-CP-CG1 is the same as the TSAS-MP-CG1 except that the objective function of the master problem is different. The model for MP-TSAS-CP-CG1 is as follows:

Minimize

$$\begin{aligned}
& c_1 \cdot C_{\max} + c_2 \cdot \sum_{j \in J} \sum_{k \in S_j} \sum_{u=1}^{n_j} t_{sj}^k z_{ju} + c_3 \cdot \sum_{j \in J} \sum_{u=1}^{n_j} t_{aj} z_{ju} + c_4 \cdot \sum_{j \in J} \sum_{k \in S_j} l_{sj}^k \\
& + c_5 \cdot \sum_{j \in J} l_{aj} + c_6 \cdot \sum_{j \in J} \sum_{k \in S_j} \sum_{u=1}^{n_j} t_{sj}^k z_{ju} + c_7 \cdot \sum_{j \in J} \max \{0, CA_{jn_j}^{aj} - d_j\} \\
& + c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} Q_j \cdot \left(CA_{jn_j}^{aj} - CS_{j0}^k - l_{sj}^k - \left(\sum_{u=1}^{n_j} z_{ju} + 1 \right) t_{sj}^k \right) / 2 - p_{sj}^k (Q_j + s_{j1}) / 2.
\end{aligned}$$

Subject to Constraints (24)–(41).

3.2. Column generation method 2 (TSAS-MP-CG2 and TSAS-CP-CG2)

In this section, we propose another column generation-based method for both the problems and designate them as TSAS-MP-CG2 and TSAS-CP-CG2. The problem is decomposed around the formation of sublots for each lot, which thus constitutes the subproblem. The master problem, then, determines a sequence in which to process the lots.

Definitions

- δ : A set of all generated columns.
- z_{ju}^p : p^{th} column of variable z_{ju} , $\forall j \in J, u = 1, \dots, n_j, p \in \delta$.
- $\lambda_p = \begin{cases} 1, & \text{if column } p \text{ is selected,} \\ 0, & \text{otherwise, } \forall p \in \delta. \end{cases}$

Master problem:

MP-TSAS-MP-CG2

Minimize

$C_{\max}$ subject to:

$$\sum_{p \in \delta} \lambda_p = 1 \quad (42)$$

$$C_{\max} \geq CA_{jn_j}^{aj}, \forall j \in J \quad (43)$$

$$CS_{j0}^k \geq CS_{in_i}^k + (x_{ij} - 1) \cdot \hat{M}, \forall i \neq j, i, j \in J^k, k \in S \quad (44)$$

$$CS_{j1}^k \geq CS_{j0}^k + l_{sj}^k + t_{sj}^k + p_{sj}^k \cdot s_{j1}, \forall k \in S_j, j \in J \quad (45)$$

$$CS_{ju}^k \geq CS_{j,u-1}^k + \sum_{p \in \delta} \lambda_p \cdot t_{sj}^k \cdot z_{ju}^p + p_{sj}^k \cdot s_{ju}, \forall k \in S_j, u = 2, \dots, n_j, j \in J \quad (46)$$

$$CA_{j0}^{aj} \geq CA_{in_i}^{aj} + (x_{ij} - 1) \cdot \hat{M}, \forall i \neq j, i, j \in A^k, k \in A \quad (47)$$

$$CA_{j1}^{aj} = CA_{j0}^{aj} + l_{aj} + t_{aj} + p_{aj} \cdot s_{j1}, \forall j \in J \quad (48)$$

$$CA_{ju}^{aj} = CA_{j,u-1}^{aj} + \sum_{p \in \delta} \lambda_p \cdot t_{aj} \cdot z_{ju}^p + p_{aj} \cdot s_{ju}, \forall u = 2, \dots, n_j, j \in J \quad (49)$$

$$CA_{j1}^{aj} \geq CS_{j1}^k + t_{sj}^k + l_{aj} + t_{aj} + p_{aj} \cdot s_{j1}, \forall k \in S_j, j \in J \quad (50)$$

$$CA_{ju}^{aj} \geq CS_{ju}^k + \sum_{p \in \delta} \lambda_p \cdot (t_{aj} + t_{sj}^k) \cdot z_{ju}^p + p_{aj} \cdot s_{ju}, \forall k \in S_j, u = 2, \dots, n_j, j \in J \quad (51)$$

$$x_{ij} + x_{ji} = 1, \forall i \neq j, i, j \in J \quad (52)$$

$$x_{ij} + x_{jl} + x_{li} \leq 2, \forall i \neq j \neq l, i, j, l \in J \quad (53)$$

Subproblem:

Let π , ϕ_{ju}^k , φ_{ju} , and ψ_{ju}^k be the dual variables associated with Constraint sets (42, 46), (49), and (51). The objective function is the reduced cost of the λ variable.

SP-TSAS-MP-CG2

Minimize

$$\sum_{j \in J} \sum_{u=1, \dots, n_j} \sum_{k \in S_j} \left[t_{sj}^k \cdot \phi_{ju}^k + t_{aj} \cdot \varphi_{ju} + (t_{aj} + t_{sj}^k) \cdot \psi_{ju}^k \right] z_{ju} - \pi$$

Subject to:

$$\sum_{u=1}^{n_j} s_{ju} = Q_j, \forall j \in J$$

$$s_{ju} \leq z_{ju} \cdot Q_j, \forall u = 1, \dots, n_j, j \in J$$

$$s_{ju} \geq s_{j,u+1}, \forall u = 1, \dots, n_j - 1, j \in J$$

$$s_{ju} - s_{j1} \geq Q_j (z_{ju} - 1), \forall u = 1, \dots, n_j, j \in J$$

$$s_{ju} - s_{j1} \leq Q_j (1 - z_{ju}), \forall u = 1, \dots, n_j, j \in J$$

$$s_{ju} \leq Q_j \cdot z_{ju}, \forall u = 1, \dots, n_j, j \in J$$

$$s_{ju} - s_{j1} \leq z_{ju} - \varepsilon, \forall u = 1, \dots, n_j, j \in J$$

$$s_{ju} \geq \varepsilon - \varepsilon (1 - z_{ju}), \forall u = 1, \dots, n_j, j \in J$$

$$z_{ju} \in \{0, 1\} \forall u = 1, \dots, n_j, j \in J$$

TSAS-CP-CG2 is the same as the TSAS-MP-CG2 except that the objective function of the master problem is different. The model for MP-TSAS-CP-CG2 is as follows:

Minimize

$$\begin{aligned}
& c_1 \cdot C_{\max} + c_2 \cdot \sum_{p \in \delta} \sum_{j \in J} \sum_{k \in S_j} \sum_{u=1}^{n_j} \lambda_p t_{sj}^k z_{ju}^p + c_3 \cdot \sum_{p \in \delta} \sum_{j \in J} \sum_{u=1}^{n_j} \lambda_p t_{aj} z_{ju}^p + c_4 \cdot \sum_{j \in J} \sum_{k \in S_j} l_{sj}^k \\
& + c_5 \cdot \sum_{j \in J} l_{aj} + c_6 \cdot \sum_{p \in \delta} \sum_{j \in J} \sum_{k \in S_j} \sum_{u=1}^{n_j} \lambda_p t_{sj}^k z_{ju}^p + c_7 \cdot \sum_{j \in J} \max \{0, CA_{jn_j}^{aj} - d_j\} \\
& + c_8 \cdot \sum_{j \in J} \sum_{k \in S_j} Q_j \cdot \left(CA_{jn_j}^{aj} - CS_{j0}^k - l_{sj}^k - \left(\sum_{p \in \delta} \sum_{u=1}^{n_j} \lambda_p z_{ju}^p + 1 \right) t_{sj}^k \right) / 2 - p_{sj}^k (Q_j + s_{j1}) / 2.
\end{aligned}$$

Subject to Constraints (42) - (53)

3.3. Outline of the column generation approach

- Step 1: Generate a set δ of initial columns for TSAS-MP-CG1 or TSAS-MP-CG2, respectively.
- Step 2: Solve the LP relaxation of the master problem, and determine the values of dual variables.
- Step 3: Solve the subproblem by using the values of dual variables from Step 2 to obtain the minimum reduced cost. If the reduced cost is non-negative or the number of iterations is greater than or equal to the maximum number of iterations allowed, then go to Step 4; else, append the column generated from subproblem to the master problem, and go to Step 2.
- Step 4: Solve the master problem as an integer program to optimality, and terminate.

The steps for TSAS-CP-CG1 and TSAS-CP-CG2 are exactly the same except for the use of corresponding formulations in Step 1.

4. Computational investigation

In this section, we present the results of our computational investigation about the CPU times required by direct solution of TSAS-MP-MIP and TSAS-CP-MIP by the OPL CPLEX Solver (version 12.4), and also, by

the proposed column generation methods. For the column generation methods, we imposed a maximum of 10 iterations. These methods were programmed using Visual C++ (Version 2008) and OPL CPLEX Solver (version 12.4). Besides the average CPU time, we also present the gap observed between the column generation solution and optimal solution values. The test data used is presented in Table 1.

We used four instances consisting of 7, 8, 9, and 10 lots. For the instance of 7 lots, we consider 5 subassembly machines at the first stage and 2 assembly machines at the second stage; while for the instance of 8 lots, we consider 6 subassembly machines at the first stage and 3 assembly machines at the second stage. For the bigger-sized problems, which consist of 9 and 10 lots, we consider 7 subassembly machines at Stage 1 and 3 assembly machines at Stage 2.

For each instance of number of lots, three different number of maximum sublots were used, namely, small, medium, and large. These values are 10, 15 and 20. Thus, there are a total of $4 \times 3 = 12$ instances for both makespan and cost-based models. For each instance, we used five randomly generated combinations of lot sizes, processing times, setup times, and transfer times, thereby, leading to 60 datasets. All numerical tests were executed on a Sony computer with Intel i7 Q740 CPU and 8GB DDR3 memory.

4.1. Results for TSAS-MP-MIP, TSAS-MP-CG1 and TSAS-MP-CG2

Comparison of cpu times required by direct solution of mip formulation by cplex and column generation methods

In this section, we compare performances of the TSAS-MP-MIP, TSAS-MP-CG1, and TSAS-MP-CG2 formulations for the objective of minimizing the makespan. Table 2 presents the minimum, average and maximum values of CPU times required by direct application of CPLEX, TSAS-MP-CG1, and TSAS-MP-CG2 for 12 different instances grouped by different number of jobs and maximum number of sublots. The smallest CPU times are highlighted. Note that in 10 of the 12 instances, TSAS-MP-CG1 requires the smallest CPU time on average, while TSAS-MP-CG2 requires the smallest time in 2 instances on average. The column generation methods dominate direct solution of the MIP model by CPLEX in view of the CPU times required for all 12 instances. For example, the instances 5, 7, 9, 10, 11, and 12 cannot be solved directly by CPLEX in 3600 s of CPU time. However, both TSAS-MP-CG1 and TSAS-MP-CG2 could obtain solutions within 2001 s. On average, TSAS-MP-MIP requires 1042.8 s of CPU time, while TSAS-MP-CG1 and TSAS-MP-CG2 require 20.3 and 157.8 s of CPU time, respectively.

Performance comparison between TSAS-MP-CG1 and TSAS-MP-CG2

Since the column generation methods may not obtain optimal solutions for all the instances, we present values of optimality gap between optimal solutions and those obtained by TSAS-MP-CG1 and TSAS-MP-CG2. These are shown in Tables 3. The optimality gap is obtained as follows:

Table 1
Data used for computational investigation.

Number of jobs (J)	7, 8, 9, and 10	
Maximum number of sublots (n_j)	High	20
	Medium	15
	Low	10
Number of products in a lot (Q_j)	Uniform distribution [100,2000]	
Lot-attached setup time (ls_j^k, la_j)	Uniform distribution [5.0,10.0]	
Sublot-attached setup time (ts_j^k, ta_j)	Uniform distribution [0.1,2.0]	
Processing time per item (ps_j^k, pa_j)	Uniform distribution [0.1,2.0]	
Transfer time per sublot (t_j^k)	Uniform distribution [5.0,10.0]	
Cost coefficient ($c_1, c_2, c_3, c_4, c_5, c_6, c_7$)	Uniform distribution [0.1,1.0]	
Cost coefficient (c_8)	Uniform distribution [0.001,0.01]	

$$\text{Optimality Gap} = \frac{\text{Solution Value} - \text{Optimal Value}}{\text{Optimal Value}}$$

The smaller optimality gap values are highlighted in Tables 3. Note that for instances 1, 3, 6, and 7, TSAS-MP-CG1 obtains optimal solutions in the minimum CPU times, while for instances 3, 4, 5, and 7, TSAS-MP-CG2 obtains optimal solutions in the minimum CPU times. The largest gap values between optimal solutions and those obtained by our two column generation methods are within 8.3% for TSAS-MP-CG1 and within 1.9% for TSAS-MP-CG2. The results of CPU times and optimality gap values are also depicted graphically in Figs. 4 and 5. (Note that instance 12 is omitted from Fig. 5 since its optimal solutions are not available.) Both TSAS-MP-CG1 and TSAS-MP-CG2 were found to perform quite well. Better optimality gap values are obtained for TSAS-MP-CG2, while TSAS-MP-CG1 requires smaller CPU times, in particular, 20.3 s on average, against 157.8 s for TSAS-MP-CG2.

4.2. Results for TSAS-CP-MIP, TSAS-CP-CG1 and TSAS-CP-CG2

Comparison of cpu times required by mip and column generation methods

As in the last section, we compare the values of CPU times required for the direct solution by CPLEX of model TSAS-CP-MIP, and by TSAS-CP-CG1 and TSAS-CP-CG2. The results are presented in Table 4 and the smallest CPU times are highlighted. Since the cost-based model is much more complex than the makespan model, TSAS-CP-MIP could not solve 6 out of 12 instances within the specified time limit of 3600 s for all 5 datasets. TSAS-CP-CG2 performs better than TSAS-CP-CG1 for 8 out of 12 instances on average, while TSAS-CP-CG1 performs better for 4 out of 12 instances on average. The results of CPU times are depicted graphically in Fig. 6. The average CPU time required by direct solution by CPLEX is 2678.5 s, while TSAS-CP-CG1 and TSAS-CP-CG2 require 860.5 and 412.3 s on average, respectively.

Comparison of optimality gaps obtained for TSAS-CP-CG1 and TSAS-CP-CG2

Next, we study the optimality gap values between optimal solution values and those obtained for TSAS-CP-CG1 and TSAS-CP-CG2. These are shown in Tables 5. For the cost-based model, TSAS-CP-CG2 dominates TSAS-CP-CG1 for all instances. The optimality gap values could be obtained only for six out of twelve instances, since the other instances could not be solved to optimality. Therefore, Table 5 depicts values for only these six instances. A comparison of the available optimality gap values for TSAS-CP-CG1 and TSAS-CP-CG2 for these instances is displayed in Fig. 7. Note that the worst optimality gap values for TSAS-CP-CG2 is within 1.8%, while that for model TSAS-CP-CG1 is above 6%. Therefore, the TSAS-CP-CG2 method is much more suitable for the cost-based objective function.

Based on our experimental investigation, we observe that performing the lot sizing decisions prior to sequencing of lots (CG2 Scheme) gives rise to better quality solutions than the other way around (CG1 Scheme). Determination of lot sizes tends to decrease the processing time of a lot, which in turn, can impact sequencing of lots in obtaining a lower makespan value or cost. However, if lot sequencing is accomplished first, an adjustment in sizes or number of lots may not impact the objective function value that much because it may lead to excessive idle time.

5. Managerial insights

Our experimental investigation has revealed TSAS-MP and TSAS-CP to be suitable models to effectively manage production and delivery of customized products for a two-stage assembly system. Both the column generation schemes CG1 and CG2 provide an optimal or near-optimal solution within an hour of CPU time for the problem instances considered, for both time-based and cost-based objective functions. The solution obtained can be easily used to identify bottleneck machine(s),

Table 2

CPU times (in seconds) required for direct solution of TSAS-MP-MIP by CPLEX and by TSAS-MP-CG1 and TSAS-MP-CG2.

Instance	Lots	No of machines at Stage 1	No of machines at Stage 2	Maximum No of sublots	CPU Times(in seconds)			TSAS-MP-CG1			TSAS-MP-CG2		
					TSAS-MP-MIP Min	Ave	Max	Min	Ave	Max	Min	Ave	Max
1	7	5	2	10	5.8	8.098	11.9	0.791	1.496	3.184	0.843	2.0326	3.06
2	7	5	2	15	7.04	17.966	32.54	1.469	2.351	3.837	1.667	3.1028	4.05
3	7	5	2	20	2.51	8.334	20.62	1.608	2.9636	4.075	1.565	2.1886	4.098
4	8	6	3	10	38.92	85.938	154.44	2.21	4.3516	6.716	7.958	26.379	49.511
5	8	6	3	15	11.28	741.78	3600†	1.483	4.2552	8.393	2.532	12.5384	45.98
6	8	6	3	20	17.63	283.794	785.64	5.596	18.7506	53.999	2.601	16.3486	60.302
7	9	7	3	10	50.61	1185.906	3600†	3.022	10.5212	16.467	8.462	147.2094	520.924
8	9	7	3	15	74.1	552.218	1325.46	4.092	7.2084	13.141	8.246	36.1634	70.821
9	9	7	3	20	1482.01	2036.902	3600†	4.422	14.8536	33.124	16.639	45.8802	76.16
10	10	7	3	10	12.2	2530.686	3600†	1.649	12.5442	44.031	4.311	305.8278	719.183
11	10	7	3	15	299.72	1461.604	3600†	6.576	127.0188	487.468	45.83	324.933	1247.59
12	10	7	3	20	3600†	3600†	3600†	23.446	37.099	71.292	77.42	970.993	2000.51

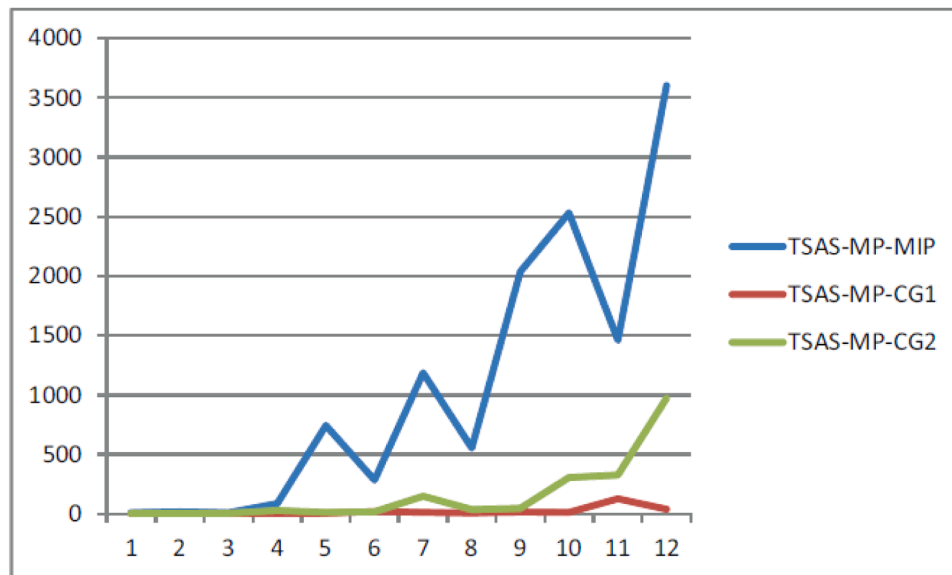
3600†: Exceeds pre-specified time limit of 3600 s.

Table 3

Comparison between optimality gap values for the solutions obtained by TSAS-MP-CG1 and TSAS-MP-CG2.

Instance	Lots	No of machines at Stage 1	No of machines at Stage 2	Maximum No of sublots	Optimality Gap TSAS-MP-CG1			TSAS-MP-CG2		
					Min	Ave	Max	Min	Ave	Max
1	7	5	2	10	0.000%	1.153%	4.447%	0.118%	0.237%	0.414%
2	7	5	2	15	0.019%	2.018%	6.154%	0.090%	0.555%	0.974%
3	7	5	2	20	0.000%	1.188%	4.652%	0.000%	0.816%	1.862%
4	8	6	3	10	0.009%	0.397%	1.352%	0.000%	0.431%	0.975%
5	8	6	3	15	0.578%	4.113%	6.989%	0.000%	0.341%	0.591%
6	8	6	3	20	0.000%	2.439%	5.805%	0.122%	0.263%	0.378%
7	9	7	3	10	0.000%	2.750%	8.227%	0.000%	0.146%	0.401%
8	9	7	3	15	0.003%	1.199%	4.353%	0.014%	0.320%	0.961%
9	9	7	3	20	0.113%	2.192%	5.700%	0.104%	0.194%	0.339%
10	10	7	3	10	0.069%	0.072%	0.076%	0.006%	0.026%	0.047%
11	10	7	3	15	0.000%	2.005%	7.978%	0.190%	0.905%	1.451%
12	10	7	3	20	N/A	N/A	N/A	N/A	N/A	N/A

N/A: Not available since optimal solution cannot be obtained.

**Fig. 4.** Average CPU times (in seconds) required by TSAS-MP-MIP, TSAS-MP-CG1 and TSAS-MP-CG2.

which can help in further improving system performance by reducing processing and/or setup times on these machines. We have further investigated this aspect below.

Consider a scenario consisting of 7 lots with 5 subassembly machines at the first stage and 2 assembly machines at the second stage. We have

investigated the impact of the following parameters on system performance: Parameter 1. Processing time per unit on the bottleneck machine at Stage 1; Parameter 2. Sublot-attached setup time on the bottleneck machine at Stage 1; Parameter 3. Transfer time of a sublot from the bottleneck machine at Stage 1 to an assembly machine at Stage 2;

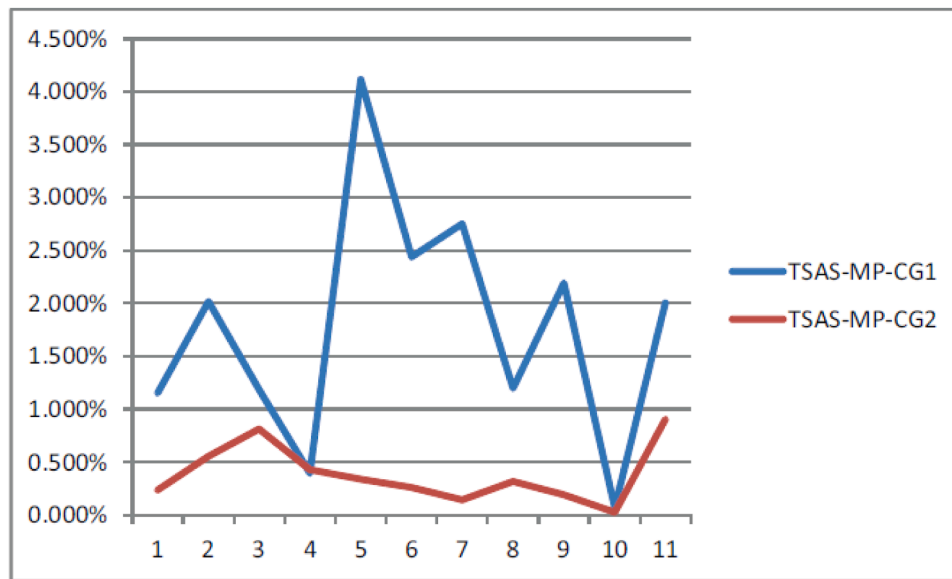


Fig. 5. Average optimality gap values for the solutions obtained by TSAS-MP-CG1 and TSAS-MP-CG2.

Table 4

CPU times (in seconds) required for direct solution of TSAS-CP-MIP by CPLEX and by TSAS-CP-CG1 and TSAS-CP-CG2.

Instance	Lots	No of machines at Stage 1	No of machines at Stage 2	Maximum No of sublots	CPU Times(in seconds) TSAS-CP-MIP			TSAS-CP-CG1			TSAS-CP-CG2		
					Min	Ave	Max	Min	Ave	Max	Min	Ave	Max
1	7	5	2	10	12	29.816	47.07	0.978	2.6806	7.018	3.416	9.3748	25.431
2	7	5	2	15	37.49	1017.352	3600†	2.446	340.743	1651.03	4.02	15.6112	24.734
3	7	5	2	20	299.32	1393.41	3600†	26.594	423.0006	1312.09	7.993	23.9448	51.519
4	8	6	3	10	919.36	2714.434	3600†	3.906	36.5668	90.51	3.701	17.7648	34.879
5	8	6	3	15	1553.24	2781.848	3600†	43.77	881.834	3600†	9.596	32.7656	86.831
6	8	6	3	20	309.55	2605.226	3600†	29.828	1101.2468	3600†	21.897	50.4528	105.427
7	9	7	3	10	3600†	3600†	3600†	1.336	3.3268	5.025	4.955	30.9434	58.207
8	9	7	3	15	3600†	3600†	3600†	72.649	1538.2102	3600†	13.11	95.3766	332.86
9	9	7	3	20	3600†	3600†	3600†	116.499	2213.5954	3600†	29.63	984.3202	3600†
10	10	7	3	10	3600†	3600†	3600†	10.815	119.1352	521.46	67.444	377.7882	767.873
11	10	7	3	15	3600†	3600†	3600†	5.991	799.3024	3600†	154.389	1441.9346	3600†
12	10	7	3	20	3600†	3600†	3600†	180.199	2865.8078	3600†	365.079	1867.3334	3600†

3600†: Exceeds pre-specified time limit of 3600 s.

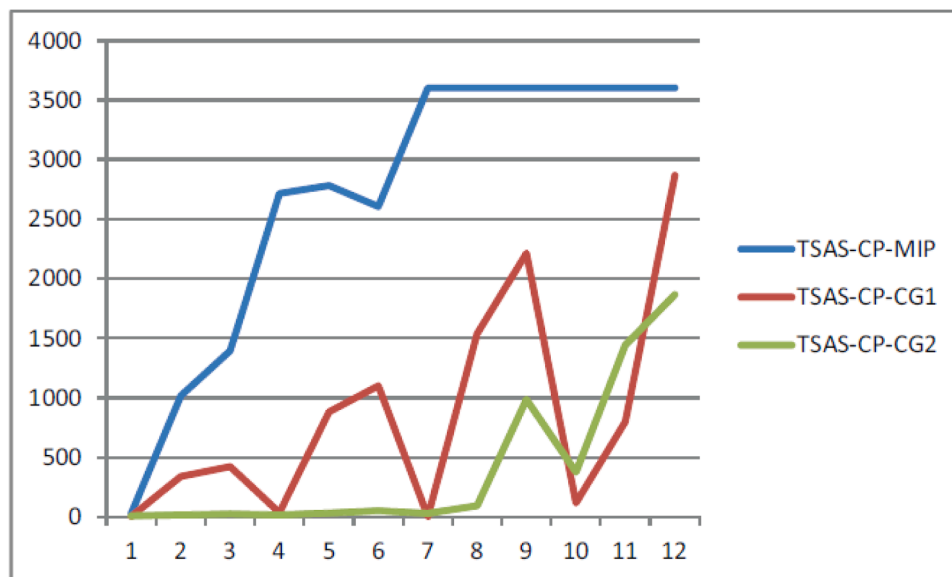
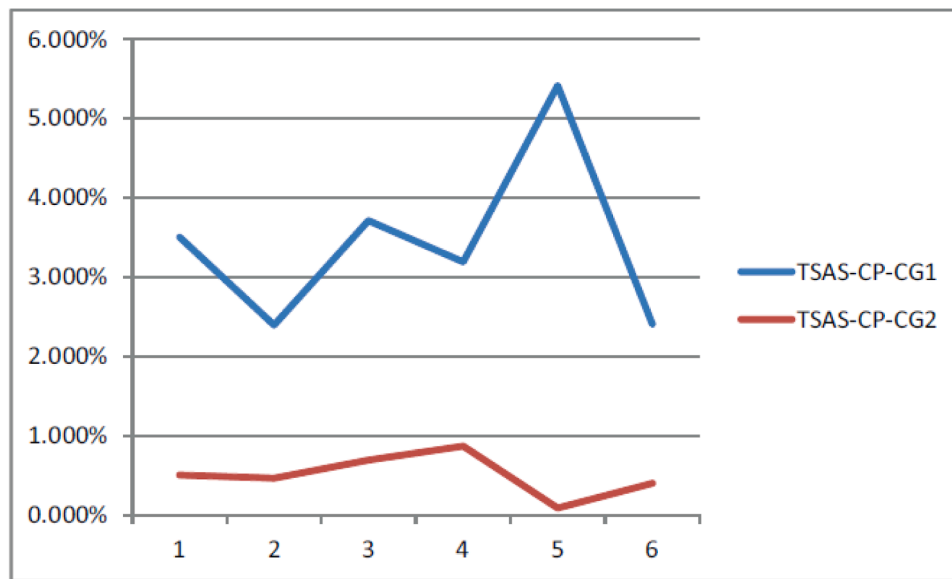


Fig. 6. Average CPU times (in seconds) required by TSAS-CP-MIP, TSAS-CP-CG1 and TSAS-CP-CG2.

Table 5

Comparison between optimality gap values for the solutions obtained by TSAS-CP-CG1 and TSAS-CP-CG2.

Instance	Lots	No of machines at Stage 1	No of machines at Stage 2	Maximum No of sublots	Optimality Gap TSAS-CP-CG1			TSAS-CP-CG2		
					Min	Ave	Max	Min	Ave	Max
1	7	5	2	10	0.556%	3.502%	6.250%	0.062%	0.505%	1.255%
2	7	5	2	15	1.981%	2.395%	3.146%	0.108%	0.464%	0.974%
3	7	5	2	20	2.706%	3.711%	4.915%	0.221%	0.694%	1.233%
4	8	6	3	10	1.058%	3.190%	5.323%	0.036%	0.871%	1.705%
5	8	6	3	15	5.408%	5.414%	5.420%	0.045%	0.087%	0.130%
6	8	6	3	20	1.240%	2.408%	3.575%	0.398%	0.402%	0.406%

**Fig. 7.** Comparison of average optimality gap values for the solutions obtained by TSAS-CP-CG1 and TSAS-CP-CG2.**Table 6**

Impact of a reduction in parameter values on the makespan value.

Parameter(s)	0%	–5%	–10%	–15%	–20%
1	606.6	–3.16%	–3.16%	–3.21%	–3.21%
2	606.6	–0.18%	–0.37%	–0.56%	–0.75%
3	606.6	–0.04%	–0.08%	–0.12%	–0.16%
1, 2, and 3	606.6	–3.16%	–3.16%	–3.21%	–3.21%
4	606.6	–0.10%	–0.20%	–0.31%	–0.41%
5	606.6	–0.02%	–0.04%	–0.06%	–0.08%
4 and 5	606.6	–0.12%	–0.24%	–0.37%	–0.49%
1, 2, 3, 4, and 5	606.6	–3.25%	–3.35%	–3.46%	–3.56%

Table 7

Impact of a reduction in parameter values on total cost.

Parameter(s)	0%	–5%	–10%	–15%	–20%
1	1837.96	–0.78%	–1.49%	–1.84%	–1.96%
2	1837.96	–0.02%	–0.04%	–0.06%	–0.08%
3	1837.96	–0.13%	–0.26%	–0.40%	–0.53%
1, 2, and 3	1837.96	–0.96%	–1.86%	–2.29%	–2.57%
4	1837.96	–0.44%	–0.88%	–1.33%	–1.74%
5	1837.96	–0.01%	–0.03%	–0.05%	–0.06%
4 and 5	1837.96	–0.45%	–0.91%	–1.37%	–1.79%
1, 2, 3, 4, and 5	1837.96	–1.41%	–2.77%	–3.66%	–4.39%

Parameter 4. Processing time per unit on the bottleneck machine at Stage 2; Parameter 5. Sublot-attached setup time on the bottleneck machine at Stage 2. The results on the change in the makespan value are presented in Table 6, while those for the total cost are presented in Table 7. The parameter values are reduced from 5% to 20% in steps of

5%. Referring to Table 6, note that, a reduction in the values of parameters at Stage 1 has a greater influence on reducing makespan value than a corresponding reduction of Stage 2 parameter values. For instance, a 5% reduction in the values of parameters at Stage 1 causes a 3.16% reduction in makespan value while the same level of reduction in the values of Stage 2 parameters results in only 0.12% reduction in makespan value, which highlights the importance of making earlier decisions. Thus, if the management needs to further improve performance, it is better to invest in the first stage processes than those at the second stage. Also, note that, a greater amount of reduction in makespan value is achieved by a reduction in per unit processing time than per unit setup time. This is mainly because the size of a lot is greater than the number of setups. Also, it is interesting to note that a smaller level of reduction in per unit processing time causes the greatest reduction in makespan value after which the percentage reduction slows down (see Fig. 8). In other words, the minimum makespan value does not decrease linearly with reduction in per unit processing time, which is mainly because of the overlapping nature of the processing of the lots on the machines.

Referring to Table 7, the impact of a reduction in parameter values on the total cost value is a little more confounding because of the simultaneous consideration of several criteria. However, note that, as in the case of makespan value, the impact of a reduction in Stage 1 parameter values is higher on the total cost value than those of the Stage 2 parameters. The other behaviors are similar to those for the makespan case albeit less pronounced because of the interacting impacts of various criteria, which leads to averaging of their differing behaviors. Also, for the same reason, the reduction in the total cost value is much smoother and more or less linear in nature with reduction in parameter values as

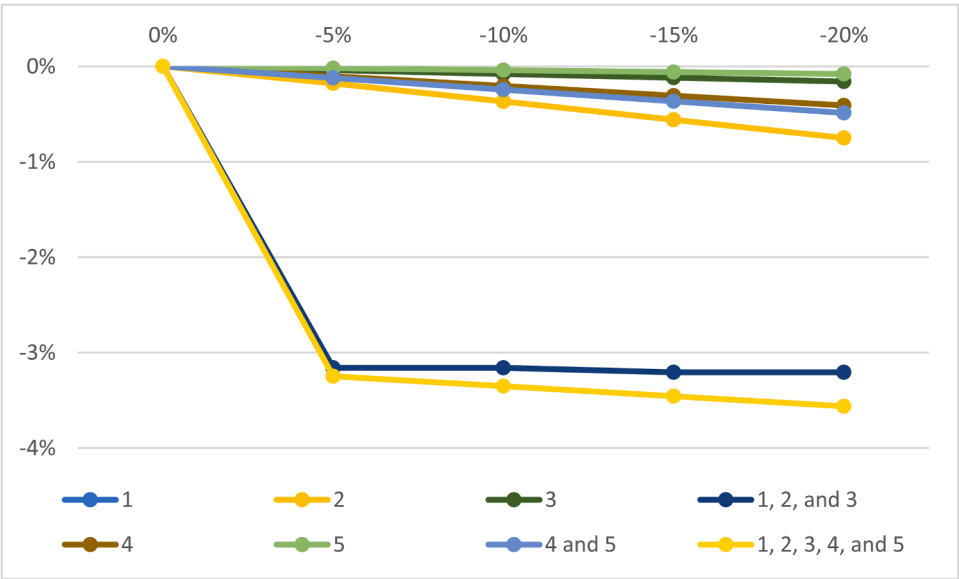


Fig. 8. Percentage change in makespan values with change in values of parameters.

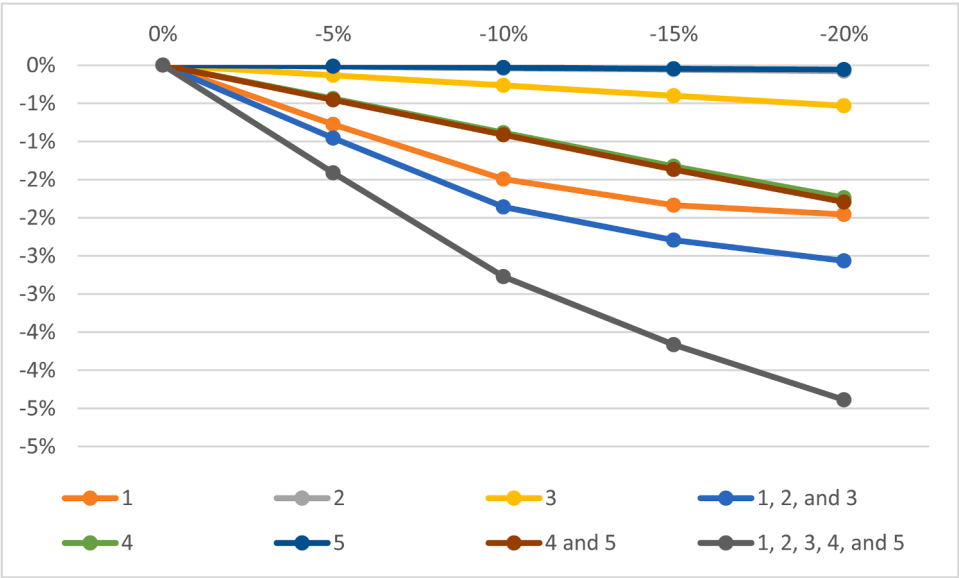


Fig. 9. Percentage change in total cost with change in values of parameters.

Table 8
Completion times of the first subplot and the entire lot (last subplot) of lots for the cost-based problem.

Time value	Lot 1	Lot 2	Lot 3	Lot 4	Lot 5	Lot 6	Lot 7
Completion time of the first subplot of a lot	93.99	112.4	211.89	272.69	363.8	469.96	573.25
Completion time of the entire lot	165.91	169.3	315.63	336.27	429.5	502.46	627.04

shown in Fig. 9. Another insight that is worth pointing out is an earlier availability of the orders for shipping in the presence of lot streaming. The completion times of the first subplot and the entire lot (last subplot) for all the lots are depicted in Table 8 for a cost-based problem. For instance, the entire Lot 1 (last subplot of Lot 1) is delivered to the customer at 165.91, while the first subplot can be delivered at 93.99, which is an improvement of 43.3% in the availability of Lot 1 to the customer.

6. Concluding remarks and ideas for future work

In this paper, we have addressed a two-stage assembly system scheduling problem in the presence of lot streaming for two objective functions: makespan and total cost. First, we developed a mixed integer programming model for this problem. Due to the difficulty of solving this model directly for large instances, we developed two column generation-based methods for each objective function type (TSAS-MP-CG1, and TSAS-MP-CG2 for makespan and TSAS-CP-CG1 and TSAS-CP-CG2 for total cost). Our computational investigation has revealed that

the proposed column generation methods generate solutions that are close to optimum. The optimality gap values are less than 8.3% (TSAS-MP-CG1) and 1.9% (TSAS-MP-CG2) for the makespan model, and are less than 6.3% (TSAS-CP-CG1) and 1.8% (TSAS-CP-CG2) for the cost-based model. Therefore, the TSAS-MP-CG2 and TSAS-CP-CG2 methods perform better for the makespan and cost-based objective functions, respectively. In addition, the column generation methods find solutions in a few seconds of CPU time while the direct solution of a mixed integer programming model of the problem by CPLEX requires much larger CPU times.

Besides developing an effective methodology for determining the number and sizes of sublots to use for processing jobs in each lot, our work has also generated some useful managerial insights, as follows: (1) a reduction in the per unit processing time and setup time values for the processors at Stage 1 has a greater impact on the objective function value (both makespan and total cost) than those for the processors at Stage 2, (2) a reduction in per unit value of processing time at the bottleneck machine(s) at Stage 1 has a greater impact on makespan or total cost value than a reduction in per unit setup time, (3) minimum makespan value does not decrease linearly with reduction in parameter values while the total cost value seems to have that behavior because of the interacting impacts of various criteria, which leads to averaging of their differing behaviors, and also, for the same reason, (4) the reduction in the total cost value is much smoother and more or less linear in nature with reduction in parameter values.

Even though, in this paper, we have addressed a lot streaming problem for a two-stage assembly system, it lays a foundation for its further development for multiple stages. Future research can also be conducted by relaxing other assumptions, like equal subplot sizes or by considering other objective functions. It would also be interesting to investigate applicability of the column generation-based methodology developed in this paper for these extensions.

Declaration of Competing Interest

We do not have any conflict of interest with the individuals on the Editorial Board of the journal or reviewers'.

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References

- Chang, J. H., & Chiu, H. N. (2005). A comprehensive review of lot streaming. *International Journal of Production Research*, 43(8), 1515–1536.
- Cheng, M., Mukherjee, N. J., & Sarin, S. C. (2013). A review of lot streaming. *International Journal of Production Research*, 51(23–24), 7023–7046.
- Cheng, M., & Sarin, S. C. (2013). Two-stage, Multiple-lot, Lot Streaming Problem for a 1 + 2 Hybrid Flow Shop. In , 46. *IFAC Proceedings Volumes. 7th IFAC Conference on Manufacturing Modelling, Management, and Control* (pp. 448–453).
- Cheng, M., Sarin, S. C., & Singh, S. (2016). Two-stage, single lot, lot streaming problem for a 1 + 2 hybrid flow shop. *Journal of Global Optimization*, 66, 263–290.
- Fern, E. J. (2003). Six steps to the future: How mass customization is changing our world.
- Fogliatto, F. S., da Silva, G. J. C., & Borenstein, D. (2012). The mass customization decade: An updated review of the literature. *International Journal of Production Economics*, 138(1), 14–25.
- Gupta, J. N. (1988). Two-stage, hybrid flow shop scheduling problem. *The Journal of the Operational Research Society*, 39(4), 359–364.
- Li, J. Q., Tao, X. R., Jia, B. X., Han, Y. Y., Liu, C., Duan, P., et al. (2020). Efficient multi-objective algorithm for the lot-streaming hybrid flow shop with variable sub-lots. *Swarm and Evolutionary Computation*, 52, Article 100600.
- Liu, J. (2008). Single-job lot streaming in m+1 two-stage hybrid flow shops. *European Journal of Operational Research*, 187(3), 1171–1183.
- Nejati, M., Mahdavi, I., Hassanzadeh, R., & Mahdavi-Amiri, N. (2016). Lot streaming in a two-stage assembly hybrid flow shop scheduling problem with a work shift constraint. *Journal of Industrial and Production Engineering*, 33(7), 459–471.
- Pine II, B. J., Victor, B., & Boynton, A. C. (1993). Making mass customization work. *Harvard Business Review*, 71(5), 108–118.
- Sarin, S. C., Sherali, H. D., & Bhootra, A. (2005). New tighter polynomial length formulations for the asymmetric traveling salesman problem with and without precedence constraints. *Operations Research Letters*, 33(1), 62–70.
- Sarin, S. C., Yao, L., & Trietsch, D. (2011). Single-batch lot streaming in a two-stage assembly system. *International Journal of Planning and Scheduling*, 1(1–2), 90–108.
- Silveira, G. D., Borenstein, D., & Fogliatto, F. (2001). Mass customization: Literature review and research directions. *International Journal of Production Economics*, 72(1), 1–13.
- Tsubone, H., Ohba, M., & Uetake, T. (1996). The impact of lot sizing and sequencing on manufacturing performance in a two-stage hybrid flow shop. *International Journal of Production Research*, 34(11), 3037–3053.
- Wang, S., Kurz, M., Mason, S. J., & Rashidi, E. (2019). Two-stage hybrid flow shop batching and lot streaming with variable sublots and sequence-dependent setups. *International Journal of Production Research*, 57(22), 6893–6907.
- Yao, L., & Sarin, S. C. (2014). Multiple-lot, lot streaming in a two-stage assembly system. *Essays in production, project planning and scheduling* (pp. 357–388). Springer.
- Zhang, B., Pan, Q., Gao, L., Zhang, X., Sang, H., & Li, J. (2017). An effective modified migrating birds optimization for hybrid flow shop scheduling problem with lot streaming. *Applied Soft Computing*, 52, 14–27.
- Zhang, W., Liu, J., & Linn, R. J. (2003). Model and heuristics for lot streaming of one job in m-1 hybrid flow shops. *International Journal of Operations and Quantitative Management*, (9), 49–64.
- Zhang, W., Yin, C., Liu, J., & Linn, R. J. (2005). Multi-job lot streaming to minimize the mean completion time in m-1 hybrid flow shops. *International Journal of Production Economics*, 96(2), 189–200.